Ex:- $S = \{(1,2,1), (2,1,-4), (3,-2,1)\}$

$v = (7,1,9)$

Sol:- $v = \alpha_1 u_1 + \alpha_2 u_2 + \alpha_3 u_3$

$$\alpha_1 = \frac{\langle v, u_1 \rangle}{\langle u_1, u_1 \rangle} = \frac{18}{6} = 3$$

$$\alpha_2 = \frac{\langle v, u_2 \rangle}{\langle u_2, u_2 \rangle} = \frac{-21}{21} = -1$$

$$\alpha_3 = \frac{\langle v, u_3 \rangle}{\langle u_3, u_3 \rangle} = \frac{28}{14} = 2$$

Prove that
→ If S is orthogonal then S is LI.

Theorem:-

Suppose $w_1, w_2, \ldots, w_r$ form an orthogonal set of non-zero vectors in V. Let v be any vector in V and we define $v' = v - (c_1 w_1 + c_2 w_2 + \ldots + c_r w_r)$. Then $v'$ is orthogonal to $w_1, w_2, \ldots, w_r$

Proof:- $\langle v', w_i \rangle$

$= \langle v - c_1 w_1 - c_2 w_2 - \ldots - c_r w_r, w_i \rangle$

$= \langle v, w_i \rangle - c_1 \langle w_1, w_i \rangle - \ldots - c_r \langle w_r, w_i \rangle$

$= \langle v, w_i \rangle - c_i \langle w_i, w_i \rangle \quad , i = 1, 2, \ldots, r$

$= \langle v, w_i \rangle - \frac{\langle v, w_i \rangle}{\langle w_i, w_i \rangle} \langle w_i, w_i \rangle \quad \left[ \because c_i = \frac{\langle v, w_i \rangle}{\langle w_i, w_i \rangle} \text{ for } i = 1, 2, \ldots, n \right]$

$= \langle v, w_i \rangle - \langle v, w_i \rangle$

$= 0$

$\Rightarrow v'$ is orthogonal to $w_1, w_2, \ldots, w_r$.

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